

Vivek Reddy Kasireddy

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Education

Worcester Polytechnic Institute | (BS) Computer Science & (BS) Robotics Engineering, GPA (3.81/4.00) | *7x Dean's List* **May 2026**

Skills

Languages: C/C++, Python, MATLAB, Java, Verilog, Java/Typescript

Software & Frameworks: SolidWorks, CREO, MATLAB, Quartus, ModelSim, Linux, Arduino, Git, OpenCV, Agile, ROS2, PyTorch

Hardware: NVIDIA Jetson Orin Nano, FPGA (Altera), Embedded Systems, SPI/LVDS Communication, ADC/PWM Control, Raspberry Pi, Arduino, PCB Design, Sensor Integration (LiDAR, Arducam), Robotics Platforms (Quadrotor, TurtleBot3), RTL Design

Publications

• *Deepak Singh**, *Hudson Kortus**, *Jahnvi Prudhivi**, *Vivek Reddy Kasireddy**, *N. Sanket* | **Lensless Aerial Navigation in Dark**, ICRA 2026 RIGOROUS Perception Workshop ([Paper Link](#))

Experience

Robotics Test Engineer | **Amazon Robotics – North Reading, MA**

June 2026 - Present

- Testing dexterous arm manipulation via teleoperation interfaces, evaluating task reliability across high-DOF manipulation sequences.
- Debugging bidirectional HRI communication protocols, mapping sensor inputs and controller command outputs to isolate failure modes and ensure deterministic arm behavior during live test cycles.
- Implemented a Python/Streamlit dashboard to track and visualize robot arm training data quality, coverage, integrated with the internal data collection pipeline.

Automation & Lab Test Engineering (Intern to Co-Op) | **Brooks Automation – Chelmsford, MA**

May. 2025 – May 2026

- Implemented a real-time FPGA control system on 2 Altera boards operating at **2 kHz**, enabling SPI communication over **2–10m links** with reliable data transfer and LVDS signaling for robust system performance and achieved **99% accuracy** in data integrity over SPI.
- Developed and debugged SPI-based communication pipelines between FPGA control units, with one device acting as the communication initiator (controller) and others responding (peripherals), ensuring precise timing alignment and reliable high-speed data transfer using ModelSim and this formed the foundation for higher-level control logic.
- Designed and integrated a custom vacuum chamber test stand using CREO, accelerating hardware validation cycles and supporting thermal stability testing of electromechanical systems, enabling continuous testing, and **ensuring <1% variance** in system performance.

Perception & Robotics Research Engineer | **Perception & Autonomous Robotics Lab ([PeAR](#))**, WPI

Aug. 2025 – April 2026

- Designed and built a **lensless vision system** for aerial robots using a **piece of scotch tape as a diffuser** to encode scene information and reconstruct images from raw sensor measurements and achieved **87% accuracy in autonomous navigation** across various scenes.
- Deployed a **lensless perception pipeline** on a quadrotor, enabling low-power **obstacle detection** in degraded visibility, while achieving **95% reconstruction accuracy** using ADMM-based optimization.

Software Test Engineer Intern | **Solgud Robotics (Startup) – Harvard University, Boston, MA**

June 2023 – Sept. 2023

- Debugged and resolved failures in SLAM, localization, improving navigation reliability across diverse environments and delivered **15 successful demos** of the robot in real-world settings like restaurants, and warehouses for basic tasks.

Projects

ROS 2-Based Autonomous Robot Navigation System – *ROS2, Linux, Python, Gazebo* ([GitHub](#))

March 2026 – April 2026

- Developed and deployed **custom path planning and navigation algorithms** for a TurtleBot3 using ROS 2, enabling autonomous exploration and goal-directed movement in unknown maze environments.

Modular Reinforcement Learning Framework– Deep Learning, OpenAi, API, Pytorch, Pandas ([GitHub](#))

Jan. 2026 – April 2026

- Built a modular reinforcement learning framework from scratch in Python (PyTorch), implementing DQN and Dyna-Q to support rapid experimentation across custom environments without reliance on external libraries (e.g., Stable-Baselines3).

Mudra AI – Real-Time ASL Gesture Translator – *TensorFlow, MediaPipe, Python, Gemini, ElevenLabs* ([GitHub](#))

November 2025

- Built a prototype translating ASL gestures to English text/voice using CV landmark extraction with ~90% accuracy in sign detection.

Traffic Sign Classification – Machine Learning – *Tensorflow, PyTorch, Numpy, Matplotlib* ([GitHub](#))

Oct. 2024 – Dec. 2024

- Implemented and optimized a CNN-based classification pipeline, tuning architecture, and training parameters to achieve 99% accuracy on a large-scale traffic sign dataset.

Industrial Serial-Arm Robot Manipulation – *MATLAB, Linux, Git, Embedded C* ([GitHub](#))

Jan 2025 – March 2025

- Designed a real-time machine vision tracking system using **OpenCV** and **MATLAB** to control a robotic manipulator for dynamic object following using trajectory generator and **inverse kinematics** solver for a 5-DOF robotic arm with 95% accuracy in control.

Escape Room Maze Runner Robot – *C/C++, Git* ([GitHub](#))

March 2024 – May 2024

- Designed and programmed a ramp-climbing robot capable of detecting April Tags with 1mm localization accuracy, achieving synchronized operations with a counterweight robot through C/C++ programming and communication protocols to enhance stability.

Autonomous Solar Panel Replacement Robot – *C/C++, Git* ([GitHub](#))

Jan. 2024 – March 2024

- Programmed an autonomous robot with a servo-actuated gripper and finite state machine (FSM) to efficiently remove and replace solar panels, incorporating line-following algorithms, IR remote control commands, achieved 85% accuracy in task completion.